

Qiskit integrates key elements including Terra, Aer, Ignis and Aqua. The IBM Qiskit framework provides execution platforms for applications in engineering, finance, machine learning, optimisation, chemistry, bioinformatics, and many other domains.

The cloud variant and lab of IBM Qiskit ensure that supercomputing applications can be executed without using physical systems on premises.

The formation and execution of circuits are possible using real IBM quantum computers on the cloud. IBM Quantum Lab enables the real-time implementation of supercomputing applications in IBM's quantum computers. The simulation of quantum hardware can be done using Qiskit programming with the customisation of devices and infrastructure.

Google Cirq provides an open source platform for quantum computing. Real-time quantum based circuits can be generated, and simulation can be done for performing supercomputing experiments. Google Cirq integrates the QSim simulator for researchers, academicians and practitioners to enable real-time engineering applications using quantum computers.

Scope for research and development

There is huge scope for research and development for scholars, academicians and practitioners in the field of quantum computing. Quantum computing can be used in domains like cryptography in blockchain, astronomical data analysis, drug design, cybersecurity, financial modelling, logistics optimisation, weather forecasting, computational chemistry, artificial intelligence, traffic optimisation, climate data analysis, and many others. All these fields need high accuracy in the minimum possible time. A comparison of results and logs from traditional and quantum systems can reveal how quantum computing enhances performance dramatically. 

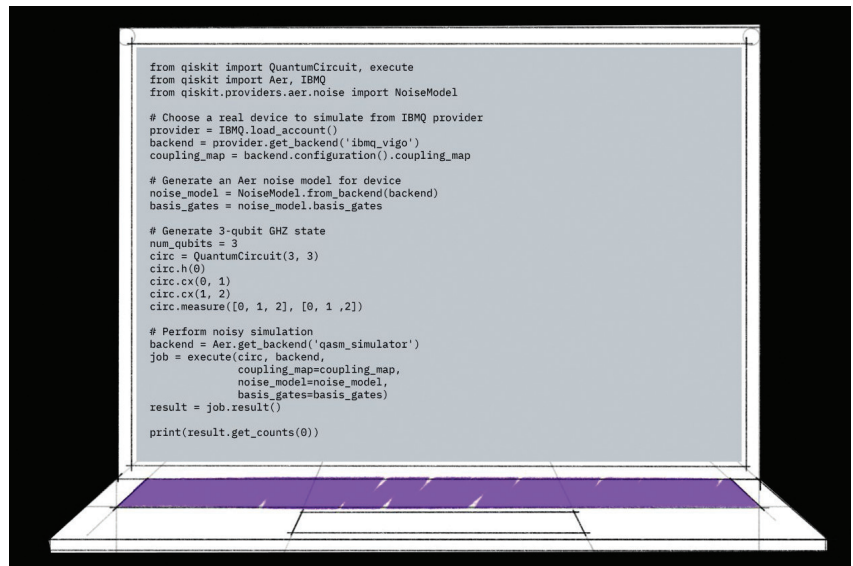


Figure 4: Quantum hardware simulation in Qiskit programming

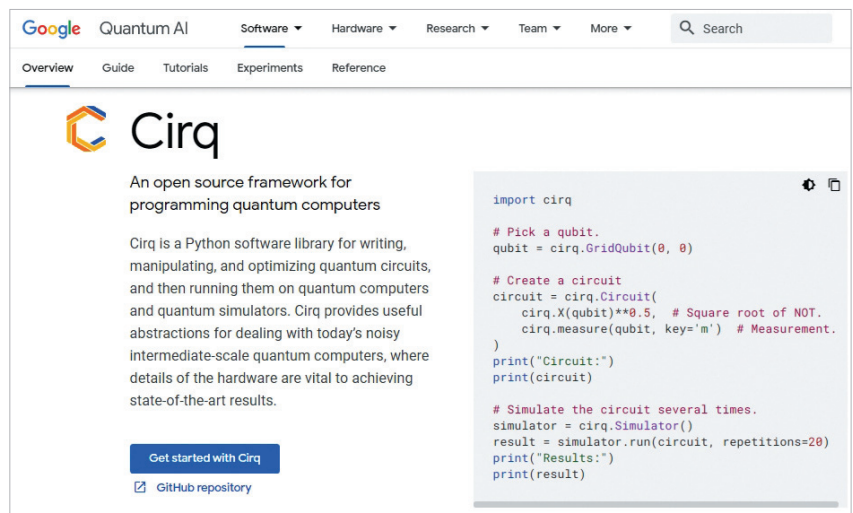


Figure 5: Google Cirq framework for quantum computers

References

- [1] David Matthews, *How To Get Started In Quantum Computing*, Nature, March 2021
- [2] IBM Qiskit, <https://qiskit.org/>
- [3] Google Cirq, <https://quantumai.google/cirq>

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